# Yulong Liu

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## Research Interests

**AI methods & discovery:** Differentiable programming, hybrid neural PDE solvers, uncertainty-aware scientific AI, and AI for Science and scientific discovery.

**Engineering:** Homogenization and multiscale modeling, coupled multiphysics simulation, and subsurface energy systems, including geothermal reservoirs and hydraulic fracturing.

## Education

**Ph.D.** in Earth Science – Cornell University 2024 – Present
*Minors: Computer Science and Scientific Computing*
**B.S.** in Mining Engineering – Northeastern University (CN) 2024

## Research Experience

**Ph.D. Candidate** – Dept. of Earth and Atmospheric Sciences, Cornell University Aug. 2024 – Present
*Advisor: Chloé Arson* *Ithaca, NY*
**Project:** Scientific AI and scientific discovery for multiscale, multiphysics geomechanics and subsurface energy systems.

- Develop coupled thermo-hydro-mechanical models in MOOSE for subsurface energy systems, including geothermal reservoirs and hydraulic fracturing.
- Study differentiable programming and hybrid neural PDE solvers, combining physics-informed networks, implicit neural representations, and neural operators with numerical simulation.
- Investigate uncertainty-aware surrogate modeling for geothermal systems, with an emphasis on uncertainty propagation and the reliability of multiphysics predictions.
- Lead a cross-institution project on PDE-grounded verification of the physical correctness of LLM-generated multiphysics simulation code.
- Study homogenization and multiscale transport in heterogeneous porous media; use AI for scientific discovery to learn interpretable effective permeability laws.

**Undergraduate Researcher** – Northeastern University (CN) Sep. 2020 – Jun. 2024
*B.S. in Mining Engineering* *Shenyang, China*
**Thesis:** A Unified Model for Microseismic Signal Classification and Arrival-Time Picking Based on Deep Learning

- Developed a unified deep-learning model for microseismic signal classification and arrival-time picking, integrating tasks typically handled in separate workflows.
- Conducted experimental and computational research on hydraulic fracturing, triaxial rock testing, and tunnel boring machines; received the Outstanding Thesis Award and National Scholarship.

## Grants, Proposals, and Computational Awards

**Learned Uncertainty-Propagation and World Models for Enhanced Geothermal Reservoirs** 2026
*National Science Foundation (NSF), ACCESS program. Student lead:* ***Yulong Liu****; Advisor: C. Arson. Total allocation and Arson group's share: 1.5M CPU hours.*

**AI-aided Computational Geomechanics** 2026
*Empire AI Alpha+Beta. PIs: Y.C. Han,* ***Yulong Liu****, A. Tristani, and C. Arson. Total allocation and Arson group's share: 5.5K GPU hours.*

**NASA FINESST Proposal – Title Withheld During Anonymous Review** 2026
*NASA ROSES-25 F.5 FINESST. Under review. Future Investigator:* ***Yulong Liu****; PI and faculty mentor: C. Arson.*

## Publications

**Selected Papers**

- **Yulong Liu**, Jonah Botvinick-Greenhouse, Yunan Yang, Chloé Arson. *Operator Learning Surrogate Modeling of Hydraulically Fractured Geothermal Injection–Production Systems: A Cornell Case Study.* ARMA US Rock Mechanics/Geomechanics Symposium, Tucson, AZ, 2026.
- **Yulong Liu**, Chloé Arson. *A Physics-Informed Neural Network for Pressurized Cavities of Arbitrary Shape in Heterogeneous Rock. Rock Mechanics and Rock Engineering* (2026). doi:10.1007/s00603-026-05882-5.
- M. Belachew, **Yulong Liu**, J. D. Frost, Chloé Arson. *Numerical Assessment of Plasticity Development and Energy Expenditure of Ant-Like Microtunnelling. Tunnelling and Underground Space Technology*, 172, 107501, 2026.
- **Yulong Liu**[†], Chloé Arson. *Physics-Informed Neural Network Surrogate Modeling of Pressurized Cavity in Homogeneous and Bilayered Media.* ARMA US Rock Mechanics/Geomechanics Symposium, D022S018R006, 2025.
- Zhenghan Song, **Yulong Liu**[*,†], et al. *Your Simulation Runs but Solves the Wrong Physics: PDE-Grounded Intent Verification for LLM-Generated Multiphysics Simulation Code.* arXiv:2605.09360. Under review at NeurIPS 2026.
- Zhenghan Song, Yunyi Li, **Yulong Liu**[†]. *Prefix-Safe Bayesian Belief Tracking for LLM Reasoning Reliability: Separating Calibration from Ranking.* arXiv:2605.27712. Under review at ACL ARR 2026.

**Thesis**

- **Yulong Liu**. *A Unified Model for Microseismic Signal Classification and Arrival-Time Picking Based on Deep Learning.* Undergraduate thesis, Northeastern University, 2024.

[*] Co-first author [†] Corresponding author

## Conferences and Presentations

**Oral presentation, 60th US Rock Mechanics/Geomechanics Symposium (ARMA)** Jun. 2026
*American Rock Mechanics Association, Tucson, AZ, USA*

**Invited talk, 20th Phase Field Methods Workshop** May 2026
*Center for Hierarchical Materials Design, Northwestern University, Evanston, IL, USA*

**Poster presentation, Cornell CEE Graduate Research Symposium** Apr. 2026
*Cornell University, Ithaca, NY, USA*

**Invited seminar: "Physics-Informed AI Optimization of Heterogeneous Rock Systems"** Dec. 2025
*Southwest Petroleum University seminar series, Chengdu, China*

**Poster presentation, 59th US Rock Mechanics/Geomechanics Symposium (ARMA)** Jun. 2025
*American Rock Mechanics Association, Santa Fe, NM, USA*

## Fellowships and Awards

**Estwing Hammer Award** – Most Outstanding EAS Graduate Student of 2024, Cornell University 2025
**Cornell University Travel Grant**, Cornell University 2025, 2026
**Outstanding Thesis**, Northeastern University 2024
**National Scholarship**, Ministry of Education, China 2024
**First Class Scholarship**, Northeastern University 2023
**Autumn Scholarship**, Northeastern University 2022
**Golden Seed Scholarship**, Northeastern University 2022

## Academic Service

- Session Co-Chair (with Wencheng Jin), GS03 Computational Geomechanics, Coupled THMC Processes, and Next-Generation Modeling, Geothermal Track, *61st U.S. Rock Mechanics/Geomechanics Symposium (ARMA 2027)*, Austin, TX, June 2027
- Reviewer, *Rock Mechanics and Rock Engineering*
- Reviewer, *Underground Space*
- Reviewer, *International Conference on Artificial Neural Networks*
- Reviewer, *Geothermal Rising Conference*

## Skills

**Programming & Scientific Computing:** Python, C++, R, MOOSE finite element framework

**Modeling & Simulation:** finite element modeling, coupled multiphysics simulation, poromechanics, computational geomechanics, CAD (e.g., AutoCAD, SolidWorks)

**Subsurface & Reservoir:** geothermal reservoirs, hydraulic fracturing, injection–production modeling, homogenization and multiscale transport in porous media

**Scientific Machine Learning:** differentiable programming, hybrid neural PDE solvers, uncertainty-aware scientific AI, AI for Science and scientific discovery; physics-informed networks, neural operators, and implicit neural representations

**LLMs for Simulation:** using large language models to accelerate simulation-software workflows and to verify the physical correctness of generated simulation code

**Experimental Methods:** rock mechanics testing (e.g., triaxial compression), laboratory instrumentation

**Currently Learning:** phase-field modeling (fracture/multiphysics), multiphase flow in porous media

**Scientific Writing:** LaTeX / Overleaf

## Professional Memberships

- Member, American Rock Mechanics Association (ARMA), 2025–Present
- Member, International Society for Rock Mechanics and Rock Engineering (ISRM), 2025–Present
- Member, Chinese Society for Rock Mechanics and Engineering (CSRME), 2024–Present
- Member, Geothermal Rising, 2025–Present